EcoLoop: Honest Builder Speaking Pointers

Quick reference cheat sheet for the updated 15-slide deck.

Swami Sant Dass Public School
github.com/Harman209/ecoloop

SLIDE 01

Title: EcoLoop

- Most school eco projects are posters or picking up bottles; we built real things.
- Two distinct halves: physical materials + zero-cost digital architecture.
- Origin: Swami Sant Dass Public School, Jalandhar.

SLIDE 02

The Packaging Trap: Multi-Layer Plastics

- Bottles are easy; multi-layer snack wrappers (MLP) are the real nightmare.
- Aluminum foil is bonded between plastic films; recyclers cannot separate profitably.
- 100% of snack packaging ends up burned or buried in dumps.

SLIDE 03

The Physical Loop

- 5 student-made products in school: Seed Paper, Bioenzymes, Uniform Bags, Eco-Bricks, Planters.
- 1 outsourced heavy material: MLP Sheets heat-pressed by our partner NGO with hydraulic machinery.
- Combines classroom crafts with external NGO capabilities into one circular loop.

SLIDE 04

how it started vs how it's going

- Initial expectation: teachers expected simple litter pickup and colored dustbins.
- My role: uniting scattered crafts under an umbrella circular brand & platform.
- Transition: moving from classroom crafts to writing software.

SLIDE 05

Why Build a Digital Platform?

- Without engineering specs, recycled products look like school crafts.
- Website publishes density, dimensions, thermal ratings, and lead times.
- Takes classroom work and presents it professionally to outside buyers.

SLIDE 06

Zero-Cost Backend on Google Sheets

- Core constraint: zero dollars, no corporate credit cards.
- Pipeline: React form → Google Apps Script webhook → Google Sheet.
- Result: $0.00/mo operating cost, zero maintenance, 99.9% uptime.

SLIDE 07

why our database bill is literally $0

- Industry contrast: paying $50/mo cloud databases vs free Google Sheets.
- We capture all production inquiries on a free Google Sheet without paying a penny.

SLIDE 08

The Inquiry Engine: Honest Review

- Candor: built in a single night sprint before an exhibition.
- Basic MVP: pre-populates specs, validates fields, logs rows to sheet.
- Plenty of room to improve (stock counts, tracking), but works for $0.

SLIDE 09

Building It Without Knowing React

- Honest truth: I don't know React or TypeScript deeply.
- Traditional hurdle: getting blocked by syntax and hours on Stack Overflow.
- Developer as architect: define rules and models; let AI agents build the code.

SLIDE 10

ChatGPT vs Agent Workspace

- ChatGPT: copy-pasting snippets back and forth, loses project context.
- Antigravity / OpenCode: agent reads files, edits across the repo, runs terminal.
- Contrast: chatting about code vs having an agent build in your workspace.

SLIDE 11

How Working with an Agent Looks

- 1. Write spec: plain English rules, colors, and layout instructions.
- 2. Agent edits files: writes components and routes directly in your folder.
- 3. You test locally: run browser, click around, see what broke.
- 4. Agent fixes bugs: paste error log back to agent; it repairs the code.

SLIDE 12

Replicating with OpenCode

- 100% free open-source terminal agent (GitHub: anomalyco/opencode).
- Gives students the same autonomous file-editing capabilities as AGY.
- One command install: `npm install -g @opencode/cli`.

SLIDE 13

Free AI Models for Students

- Free cloud: Google Gemini (AI Studio), Groq Cloud, GitHub Models.
- 100% offline: Ollama running `qwen2.5-coder:7b` on laptop with no internet.
- Zero credit cards, zero monthly subscriptions.

SLIDE 14

deployed without testing

- Engineering reality: AI claims "all bugs resolved", but live buttons explode.
- Core lesson: prompting is 20%; testing if buttons actually work is the 80%.
- Never assume AI code works until tested manually in the browser.

SLIDE 15

Thank You & Scannable QR

- Scan QR code: opens GitHub repository with full source code and live demo.
- Take-home: 1-page setup guide lets students set up OpenCode and free models tonight.
- Open the floor for questions from students and teachers.